

# Misspecification-Robust Inference for the Incremental Fit Indices CFI and TLI under an Estimated Weight

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## Purpose

The companion notes fix the estimated-weight inference of a *single* quadratic-form criterion. `rmsea_asymptotics.tex` and the RMSEA path of `higher_order_discrepancy_misspec` treat the discrepancy  $F = r'Wr$  itself; the CRMR path treats the correlation residual sum  $G = r'V_0r$ . Both reduce to one object: a statistic  $T = N\hat{F}$ , a bias-corrected noncentrality  $\hat{\delta} = T - \text{tr}(Q\Gamma_x)$  where  $Q$  is a profile Hessian and  $\Gamma_x$  the joint NACOV of the extended first-stage vector  $x = (u, \gamma)$ , and a sampling variance  $\text{Var}(T) = N g'\Gamma_x g$  obtained by the delta method from an envelope-score gradient  $g$ .

This note opens the program's *first incremental* (two-model) index. CFI and TLI are not functions of one criterion; they compare the fitted user model to the independence baseline,

$$\text{CFI} = 1 - \frac{\max(0, T_u - df_u)}{\max(0, T_b - df_b)}, \quad \text{TLI} = \frac{T_b/df_b - T_u/df_u}{T_b/df_b - 1}. \quad (1)$$

Under an estimated weight neither  $T_u$  nor  $T_b$  is  $\chi^2$ : each is a mixture  $\sum_j \lambda_j \chi_1^2$  whose mean is the generalized degrees of freedom  $\bar{Q} = \text{tr}(Q\Gamma_x)$ , not the nominal  $df$ . The single new statistical object is the *joint* law of  $(T_u, T_b)$ . The headline is that both statistics are smooth functionals of the *same*  $x$ , so their joint covariance is one bilinear form in the *one* shared  $\Gamma_x$ :

$$\text{Cov}(T_u, T_b) = N g'_u \Gamma_x g_b, \quad (2)$$

and CFI and TLI are then a ratio delta-method on the two correlated noncentralities. No new covariance machinery is needed beyond the second gradient

$g_b$  and the second profile Hessian  $Q_b$ ; the metric  $\Gamma_x$  is shared. Three consequences follow: an honest Wald interval for CFI (Corollary 1), a TLI interval that is the CFI interval rescaled by a generalized-df ratio (Corollary 2), and a baseline-dominated simplification (Remark) under which CFI's variance collapses to the RMSEA-side variance the program already computes, divided by the squared baseline noncentrality.

## 1 Setup

Write  $x = (u, \gamma) \in \mathbb{R}^{2m}$  for the extended first-stage vector: the  $m$  polychoric sample moments  $u$  (thresholds and correlations, stacked in the `OrdinalStats` order) and the DWLS weight diagonal  $\gamma = \text{diag } \hat{\Gamma}$ ,  $W = \text{diag}(1/\gamma)$ . Let

$$\sqrt{N}(\hat{x} - x_0) \xrightarrow{d} \mathcal{N}(0, \Gamma_x),$$

with  $\Gamma_x$  the joint per-unit NACOV built from the stacked casewise influence  $[g_i \mid \text{IF}_i(\gamma)]$  exactly as in `ordinal_dwls_profile_rmsea`.  $\Gamma_x$  is a property of the data, not of any fitted model: *the user model and the baseline share it*.

For a model  $M \in \{u, b\}$  (user, baseline) the minimum-discrepancy fit profiles its parameters out of

$$F_M(x) = d_M(x)' W d_M(x), \quad d_M = \sigma_M(\hat{\theta}_M) - u,$$

giving statistic  $T_M = NF_M$ , profile Hessian  $Q_M$  (the  $2m \times 2m$  extended object of `weighted_moment_profile_rmsea_estimated_weight`), generalized df  $\bar{Q}_M = \text{tr}(Q_M \Gamma_x)$ , and bias-corrected noncentrality

$$\hat{\delta}_M = T_M - \bar{Q}_M. \tag{3}$$

The user model is the fitted CFA/SEM. The baseline is the independence model: free thresholds, all polychoric correlations fixed to zero (delta scaling fixes the variances). Its residual is  $d_b = (0; -\text{vech}_{<} \hat{R})$  (thresholds profile to zero residual under diagonal  $W$ ; the correlation block is the negative sample correlations), and  $T_b$  matches `ordinal_baseline_chi2`. Because  $\sigma_b$  is *linear* in the baseline thresholds, the baseline observed bread equals its Gauss–Newton bread and  $Q_b$  is exact.

**Envelope gradient.** Since  $\hat{\theta}_M$  minimizes  $F_M$ , the  $\hat{\theta}_M$ -movement term in  $dF_M/dx$  vanishes (envelope theorem) and the gradient is the bare profile

score, identical in form for both models:

$$g_M = \left( \frac{\partial F_M}{\partial u}, \frac{\partial F_M}{\partial \gamma} \right) = \left( -2Wd_M, -d_M^2/\gamma^2 \right) \in \mathbb{R}^{2m}, \quad (4)$$

the  $\gamma$  block coming from  $\partial W/\partial \gamma_k = -1/\gamma_k^2$  in  $F_M = \sum_k d_{M,k}^2/\gamma_k$ . Equation (4) is exactly the RMSEA envelope score, evaluated once at the user residual  $d_u$  and once at the baseline residual  $d_b$ .

## 2 Joint law of the two statistics

**Theorem 1.** *Under fixed misspecification of both models and the regularity of the single-criterion program, with  $\hat{x} = \bar{x}$  the sample first-stage vector,*

$$\sqrt{N} \begin{pmatrix} F_u - F_{0,u} \\ F_b - F_{0,b} \end{pmatrix} \xrightarrow{d} \mathcal{N} \left( 0, \begin{pmatrix} g'_u \Gamma_x g_u & g'_u \Gamma_x g_b \\ g'_b \Gamma_x g_u & g'_b \Gamma_x g_b \end{pmatrix} \right).$$

*Equivalently, with  $V_{uu} = Ng'_u \Gamma_x g_u$ ,  $V_{bb} = Ng'_b \Gamma_x g_b$ ,  $V_{ub} = Ng'_u \Gamma_x g_b$ , the statistic pair satisfies  $\text{Var}(T_u) = V_{uu}$ ,  $\text{Var}(T_b) = V_{bb}$ ,  $\text{Cov}(T_u, T_b) = V_{ub}$ , which is (2).*

*Proof.* Each  $F_M(\hat{x}) = F_{0,M} + g'_M(\hat{x} - x_0) + o_p(N^{-1/2})$  by a first-order expansion, with  $g_M$  the envelope gradient (4) (the profiled parameters contribute no first-order term). Stacking the two expansions and applying the single CLT  $\sqrt{N}(\hat{x} - x_0) \xrightarrow{d} \mathcal{N}(0, \Gamma_x)$  to the common  $\hat{x}$  gives the bivariate limit; the off-diagonal is  $g'_u \Gamma_x g_b$  because both rows read the same increment  $\hat{x} - x_0$ . Scaling by  $N$  ( $T_M = NF_M$ ) gives the stated variances.  $\square$

The mean corrections  $\bar{Q}_M$  are  $O_p(1)$  and, as in the single-criterion program, are treated as fixed in the variance (their sampling fluctuation is higher order). Hence  $\text{Var}(\hat{\delta}_u) = V_{uu}$ ,  $\text{Var}(\hat{\delta}_b) = V_{bb}$ ,  $\text{Cov}(\hat{\delta}_u, \hat{\delta}_b) = V_{ub}$ .

## 3 CFI

In the regime where both models misfit,  $\hat{\delta}_u > 0$  and  $\hat{\delta}_b > 0$ , the truncations in (1) are inactive and, writing the generalized-df noncentralities for  $df$ ,

$$\widehat{\text{CFI}} = 1 - r, \quad r := \frac{\hat{\delta}_u}{\hat{\delta}_b}. \quad (5)$$

The population target is  $\text{CFI}_0 = 1 - \delta_{0,u}/\delta_{0,b} = 1 - F_{0,u}/F_{0,b}$  (the  $N$  cancels in the ratio).

**Corollary 1.** *The ratio gradient is  $\nabla r = \hat{\delta}_b^{-1}(1, -r)$ , so by the delta method*

$$\widehat{\text{Var}}(\widehat{\text{CFI}}) = \widehat{\text{Var}}(r) = \frac{1}{\hat{\delta}_b^2} \left( V_{uu} - 2r V_{ub} + r^2 V_{bb} \right), \quad (6)$$

*and a two-sided  $1 - \alpha$  interval is  $\widehat{\text{CFI}} \pm z_{1-\alpha/2} \widehat{\text{Var}}(\widehat{\text{CFI}})^{1/2}$ , clipped to  $[0, 1]$ .*

## 4 TLI

TLI is the same comparison taken per generalized degree of freedom. With  $E[T_M] \approx \bar{Q}_M + \delta_M$ , the misspecification-robust analogue of  $T/df$  is  $T_M/\bar{Q}_M \approx 1 + \delta_M/\bar{Q}_M$ , and substituting into the TLI form of (1) gives

$$\widehat{\text{TLI}} = \frac{T_b/\bar{Q}_b - T_u/\bar{Q}_u}{T_b/\bar{Q}_b - 1} = 1 - cr, \quad c := \frac{\bar{Q}_b}{\bar{Q}_u}. \quad (7)$$

Thus TLI is CFI's ratio  $r$  rescaled by the generalized-df ratio  $c$ ;  $\widehat{\text{TLI}} = 1 - (1 - \widehat{\text{CFI}}) \bar{Q}_b/\bar{Q}_u$ . Using  $\bar{Q}$  rather than the nominal  $df$  keeps TLI internally consistent with the mixture mean that defines  $\hat{\delta}$ ; when the weight is correctly specified  $\bar{Q}_M \rightarrow df_M$  and (7) recovers the lavaan point value.

**Corollary 2.** *Treating  $c$  as fixed (the  $\bar{Q}_M$  are deterministic mean corrections),*

$$\widehat{\text{TLI}} = 1 - cr \implies \widehat{\text{Var}}(\widehat{\text{TLI}}) = c^2 \widehat{\text{Var}}(\widehat{\text{CFI}}). \quad (8)$$

*The TLI interval is the CFI interval scaled by  $c = \bar{Q}_b/\bar{Q}_u$ : same  $r$ , same  $\widehat{\text{Var}}(r)$ , one extra factor. (TLI is not clipped to  $[0, 1]$ ; it can exceed 1 for  $\hat{\delta}_u < 0$ .)*

## 5 The baseline-dominated regime

Under any appreciable misspecification the independence model fits grossly, so  $\hat{\delta}_b \gg \hat{\delta}_u$ ,  $r \rightarrow 0$ , and (6) collapses to its leading term:

$$\widehat{\text{Var}}(\widehat{\text{CFI}}) = \frac{V_{uu}}{\hat{\delta}_b^2} \left( 1 - 2r \frac{V_{ub}}{V_{uu}} + r^2 \frac{V_{bb}}{V_{uu}} \right) \approx \frac{V_{uu}}{\hat{\delta}_b^2} = \frac{\text{Var}(T_u)}{\hat{\delta}_b^2}. \quad (9)$$

That is, to leading order CFI's sampling variance is the *user-model* statistic variance  $V_{uu} = Ng'_u \Gamma_x g_u$  already returned as `grad_var` by the RMSEA path, divided by the squared baseline noncentrality  $\hat{\delta}_b^2$ ; the baseline acts as a nearly constant scale. The cross-term and baseline-variance contributions are  $O(r)$

and  $O(r^2)$  corrections. The full bivariate form (6) is retained in code for honesty, and it must be: the Monte-Carlo check (`tests/checks/ordinal_cfi_inference`) finds the ratio  $(V_{uu}/\hat{\delta}_b^2)/\widehat{\text{Var}}(\widehat{\text{CFI}})$  falling from 0.99 at weak misfit to 0.35 at strong misfit, i.e. (9) is accurate only near correct specification, where “CFI inference is a rescaling of RMSEA inference” holds; as misfit grows the cross-term and baseline-variance corrections take over. The simplification is intuition, not the computation.

## 6 Estimated versus fixed weight

Zeroing the  $\gamma$  blocks of  $\Gamma_x$  (drop the  $\text{IF}(\gamma)$  rows/columns) gives the fixed-weight comparator: the  $g_M$   $\gamma$ -channel in (4) is then inert,  $V_{MM}$  and  $V_{ub}$  revert to their conventional (treat- $W$ -as-known) values, and  $\bar{Q}_M$  reverts to the fixed-weight trace. The estimated-weight law differs from current practice exactly through this channel. It enters CFI twice: through  $V_{uu}$  (the RMSEA-metric piece, where the channel is large because RMSEA’s metric *is* the weight) and through the baseline objects  $\hat{\delta}_b, V_{bb}$ . One might expect the former to dominate and CFI to inherit RMSEA’s large weight sensitivity, but empirically the *net* effect on  $\widehat{\text{Var}}(\widehat{\text{CFI}})$  is small: the Monte-Carlo check puts the  $\gamma$  share at only  $\pm 3\text{--}8\%$  (versus  $-12\%$  to  $-80\%$  for RMSEA). The ratio structure and the  $\hat{\delta}_b^2$  scaling attenuate the channel, and the  $\hat{\delta}_b$  shift under a fixed weight largely cancels it, so CFI sits at the CRMR end of the spectrum: its estimated- and fixed-weight intervals nearly coincide.

## 7 Boundary and regularity

- **User boundary.** As  $\hat{\delta}_u \downarrow 0$  (correct or near-correct user model)  $\widehat{\text{CFI}} \rightarrow 1$  sits on the parameter-space boundary, the truncation  $\max(0, \cdot)$  activates, and the normal approximation degrades. The interval is clipped at 1 and flagged. This is the RMSEA close-fit boundary made sharper by CFI’s hard ceiling.
- **Baseline boundary.** If  $\hat{\delta}_b \leq 0$  (independence model fits; essentially uncorrelated data) the ratio is undefined: report  $\widehat{\text{CFI}} = 1$ ,  $\widehat{\text{TLI}}$  and both intervals NaN, with a warning. The delta method requires  $\hat{\delta}_b$  bounded away from zero, which holds whenever the baseline is a genuine null.
- **Shared metric.**  $g'_u \Gamma_x g_b$  uses the *same*  $\Gamma_x$  for both gradients; the

implementation must assert the two profile fits returned a common  $\Gamma_x$  (they do, being data-only) before forming the cross-term.

- **TLI variance is ill-conditioned at strong misfit.** Corollary 2 treats  $c = \bar{Q}_b/\bar{Q}_u$  as fixed, but  $\bar{Q}_u$  is a signed trace that can be small (and near its sign-change cancellation) when the user model misfits strongly, giving  $c^2$  a heavy right tail. The Monte-Carlo check finds the TLI interval calibrated at weak/moderate misfit but *over-covering* at strong misfit (analytic variance  $\sim 8\times$  the empirical), i.e. conservative, not anti-conservative. The TLI *point* stays unbiased. CFI carries no such division and is the index to trust for an interval; a stabilized  $c$  for TLI is future work.
- **Single group.** As with RMSEA/CRMR, the first pass is single-group; the multi-group extension stacks  $\Gamma_x$  block-diagonally and sums the block traces, and is deferred.

## 8 Implementation map

1. User profile via `ordinal_dwls_profile_rmsea`  $\Rightarrow Q_u, \Gamma_x, T_u, \bar{Q}_u$ ; form  $g_u$  from (4) with the fitted residual  $d_u$  (as the RMSEA path already does).
2. Baseline profile from the analytic independence block: jacobian  $D_b = [I_{nth}; 0]$ , residual  $d_b$ , weight  $\gamma$ , shared  $\Gamma_x$ , Gauss-Newton bread  $D_b' W D_b$ ,  $K_b = I$ . Route through `weighted_moment_profile_rmsea_estimated_weight`  $\Rightarrow Q_b, T_b, \bar{Q}_b$ ; cross-check  $T_b$  against `ordinal_baseline_chi2`. Form  $g_b$  from (4) with  $d_b$ .
3.  $\hat{\delta}_u = T_u - \bar{Q}_u$ ,  $\hat{\delta}_b = T_b - \bar{Q}_b$ ;  $V_{uu} = N g_u' \Gamma_x g_u$ ,  $V_{bb} = N g_b' \Gamma_x g_b$ ,  $V_{ub} = N g_u' \Gamma_x g_b$ .
4. CFI (5), (6); TLI (7), (8); intervals per Corollaries 1–2 with the boundary handling of Section 7.
5. `estimated_weight=false` zeros the  $\gamma$  blocks of  $\Gamma_x$  before steps 3–4 (fixed-weight comparator).